

• QUANTIXCORE TECHNOLOGIES LLC

SVC — Security Video Conference

A fully self-hosted, encrypted video-conferencing and secure document-exchange platform for government bodies and privacy-critical organizations — ending dependence on foreign cloud services like Zoom and Google Meet.



Self-hosted



Elixir / Phoenix



LiveKit SFU



Native Android — live

● PROBLEM

Why **SVC** is needed



Foreign services are unsafe

Zoom / Meet store data on overseas servers — unsuitable for state secrets and confidential information.



Document leakage

Files sent over unprotected channels fall out of the organization's control.



No accountability

If no one records who viewed what, there is no responsibility trail.



Legal non-compliance

Conflicts with data-localization requirements for personal and state data.

● SOLUTION

One platform — full control

SVC keeps all data and infrastructure inside the organization's own perimeter, with no dependency on any foreign cloud.



Encrypted video & audio

Media encrypted in transit (WebRTC / DTLS-SRTP); end-to-end (SFrame) on the roadmap.



Secure document exchange

Encrypted storage, versioning, expiry and revocation.



Immutable audit

Every sensitive action recorded in an append-only log.



Role-based access & privacy

Strict role hierarchy; meetings visible only to their organizer and invited members.

A working product, not just an idea

SVC is already deployed and running on the public internet — a web admin panel and a native Android application, both live today.

admin.co1nlist.uz **Android**

Live web app (login & panel)

Native app — downloadable & working

24 / 7

Continuously online



svc.co1nlist.uz — landing & app download



Authentication & RBAC working



Demo video (2:47): github.com/quantixcore-technologies/security-video-conference → [docs/award/SVC_Demo_EN.mp4](#)

Who it serves & how it earns



Government bodies (GovTech)

sovereignty is the primary driver



Banks & finance

regulation, banking secrecy



Defense & law enforcement

confidential communication



Healthcare, legal, enterprise

patient / client confidentiality

Revenue model

1 Annual license + per-seat pricing

2 Deployment & integration (E-IMZO, existing systems)

3 Support & SLA contracts

4 Extensions: additional modules, custom features

● COMPETITIVE ADVANTAGE

Why SVC?

CAPABILITY	ZOOM / MEET	GENERIC ON-PREM	SVC
Self-hosted / data sovereignty	✗	⚠	✓
Native screen-capture protection	✗	✗	✓
Role hierarchy + meeting privacy	⚠	⚠	✓
Immutable audit (append-only)	✗	⚠	✓
Forensic watermark + capture policy	✗	✗	✓
Local language & support	✗	⚠	✓

Role hierarchy



Super Administrator

Full system control; registers staff and assigns roles.



HR / Administrator

Staff and department management.



Manager

Creates and runs meetings; invites equal / junior staff only.



Employee · Security Officer

Attend meetings, exchange documents / monitor security & audit.

✓ Enforced today

No self-registration — users are added only by an administrator. Meetings are private: even a super-admin cannot see another manager's meeting unless invited.

Video conferencing



1-to-1 calls

Direct encrypted communication.



Group meetings

Efficient scaling through an SFU (LiveKit).



Screen sharing

Protected with watermarking.



Adaptive quality

Automatic bitrate adjustment to network conditions.

Document **exchange**



Encrypted upload

Documents stored and transferred encrypted.



Signing workflow

Print → sign → scan return (national e-signature integration planned).



Expiry & revocation

Set access windows or revoke instantly.



Versioning & audit

Every version kept; each open is logged.

● SECURITY

Layered **protection**



Encryption in transit

WebRTC media over DTLS-SRTP; end-to-end (SFrame) on the roadmap.



Hardened perimeter

Cloudflare Tunnel — server IP hidden, ports closed, DDoS protection.



Immutable audit

Every action in an append-only log.



Screen-capture protection

Android FLAG_SECURE blocks screenshots & screen-recording in calls.

External recording: a **realistic answer**

⚠ A physical limit

No software or AI can fully block someone recording the screen with a second phone or microphone. So we rely not on prevention alone, but on detection and traceability.

1 Dynamic watermark — identifies the leaker through a forensic trace

2 On-device capture blocking where the OS allows it (Android / Windows)

3 Organizational control — supervised rooms, device policy

System design



Clients

Native mobile (Android — live; iOS planned), Desktop (Tauri), Web admin (Phoenix LiveView).



Cloudflare Tunnel

HTTPS / WSS, hidden IP, DDoS protection.



Phoenix / Elixir backend

Auth, REST API, signaling, PubSub, Presence.



LiveKit SFU

Scalable WebRTC media routing for group calls.



PostgreSQL + encrypted files

Ecto, encrypted storage, org-scoped data.

The stack

BACKEND

Elixir · Phoenix

REAL-TIME

Channels · PubSub

WEB UI

Phoenix LiveView

VIDEO SFU

LiveKit (WebRTC)

MOBILE

Kotlin · Jetpack Compose

DATABASE

PostgreSQL · Ecto

AUTH

Argon2 · TOTP 2FA

DESKTOP

Tauri (Rust)

INFRA

Cloudflare · systemd

The BEAM VM sustains hundreds of thousands of concurrent, low-latency connections — ideal for video signaling; OTP supervision makes it fault-tolerant.

What is **already built**



Backend + live deployment

Phoenix app running at admin.co1nlist.uz, hardened.



Native Android app — delivered

Login, meetings, notifications, screen-capture protection, self-update.



RBAC & meeting privacy

Role hierarchy + confidential meetings, verified end-to-end.



Infrastructure

Cloudflare Tunnel, PostgreSQL, encrypted backups, private Git.



Early-stage: no commercial customers or revenue yet. Next: cross-device group video (LiveKit Cloud / dedicated IP), E-IMZO integration, iOS.

Development roadmap

1

MVP — done

Auth, admin panel, RBAC, native Android, live deploy.

2

Scale-up (3–6 mo)

Cross-device group video (SFU), forensic watermark, E-IMZO, richer audit.

3

Native hardening (6–9 mo)

iOS, AI second-camera detector, end-to-end (SFrame) encryption.

4

Production (9–12 mo)

Load & security audit, pilot customer → commercialization.

• QUANTIXCORE TECHNOLOGIES LLC

The team

A focused engineering team building a sovereign, secure communication platform — a working product, a clear market, an honest plan.

FOUNDER & CEO · AI ENGINEER

Shuhrat Ahmedov

DEVOPS ENGINEER

Furqatbek Yuldashov

MOBILE DEVELOPER

Yoqutjon Ilhomboyeva

FRONTEND DEVELOPER

Madamin Madaminov



President Tech Award 2026



Best Startup Project



Cybersecurity / GovTech

Thank you! · svc.co1nlist.uz ·  [demo video](#) · [source on GitHub](#)